

Project Documentation

Project Title: [AI study buddy API]

➤ 1. Introduction

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Description

StudyBuddy is an all in one, AI-powered study platform designed specifically for the rigorous education ecosystem at Georgia Tech. By bridging the gap between scattered course materials and vast Knowledge of Large Language Models (LLMs), the platform transforms static and multi-stage lecture slides, textbooks and office hours into an interactive, personalized tutor. Integrates local non-proprietary models (Llama 3, Phi-3, and Gemma 2) directly on the student's machine, ensuring data privacy and open access to all GT students. The tool provides a rich dashboard that captures the student's progress across various concepts.

Vision

The StudyBuddy is an AI-enabled platform supporting Georgia Tech student classroom experience. This nexus of AI agents allows students to personalize their classroom activities to their own learning style and pace of understanding. Through Canvas integration, students rely on AI to intensify their engagement with course content and improve their academic performance.

Release 1.0

Provides lecture summaries and course content references tailored to student questions, with automatic quiz generation for self-assessment and comprehension checks. Students can interact with the system to gain a deeper understanding of complex concepts. They can generate flashcards and other materials based on coursework materials via Canvas and beyond.

FEATURES

- **RAG with Precise Citations** – uploaded course syllabus, textbooks, or lecture slides (PDF/Text). When you ask a question, the AI doesn't just guess—it scans your library, synthesizes an answer, and provides page-specific citations
- Smart PDF Processing: Handles complex layouts found in academic slides
- Dual-Mode Chat: Switch seamlessly between a general study assistant and a document-specific research tool
- **Active Recall Engines** – transforms content into active learning sessions:
- Practice Generator: Instantly create quizzes with variable difficulty levels based on uploaded files.
- Flashcard Builder: Auto-generate decks from dense text and export them to CSV for use in spaced-repetition tools (Anki).
- **Campus Integration**
- Canvas Sync: Connect directly to your Georgia Tech Canvas account via API key. The system ingests your assignments and calendar to keep you organized.
- Study Planner: dynamic weekly planning with assignment tracking to prevent cramming.
 - Whether "Study Buddy AI" is free depends entirely on which specific app or platform you are using, as many different tools share this common name.
- **Study Buddy AI:** Offers basic learning features, but exact pricing depends on the specific tools and limits you access.
- **Mobile Apps (e.g., Study Buddy: AI Problem Solver):** Frequently offer a short free trial (such as 3 days) before requiring a weekly subscription.
- **Other Platforms (e.g., StudyFetch, Study AI Buddy):** Typically provide a limited free trial or a single free study pack, requiring a paid upgrade for unlimited

- To create a free API key, you must sign up for a provider that offers a free tier (such as Google, OpenRouter, or GitHub) and generate a token through their developer dashboard. Because "API key" is a general term, the exact steps depend entirely on the service or AI model you want to access.
- Below are the step-by-step guides for the most popular platforms providing free API keys.
- **Top AI Study Buddies by Use Case**
 - **Best Overall:** ChatGPT excels at conversational tutoring, breaking down complex topics, quizzing you on weak spots, and going line-by-line through practice problems.
 - **Best for Your Own Notes & PDFs:** Google Gemini and NotebookLM turn your uploaded lecture notes, textbooks, and syllabi into grounded summaries, audio overviews, and personalized quizzes.
 - **Best for Essay Writing and Deep Analysis:** Claude handles massive blocks of text and provides nuanced reasoning for literature and humanities.
 - **Best for Research and Citations:** Perplexity acts as a fast research assistant by pulling verified, real-time web sources and academic citations
 - **Best for Memorization:** Tools like Quizlet use active recall and AI-generated flashcards to help you retain formulas, definitions, and vocabular

1. Google Gemini API Key (Most Common for Study Buddies)

- Go to Google AI Studio.
- Sign in with your Google account.
- Click **Get API Key** or **Create API Key** in the sidebar.
- Copy your unique key and save it securely. Set it as an environment variable named

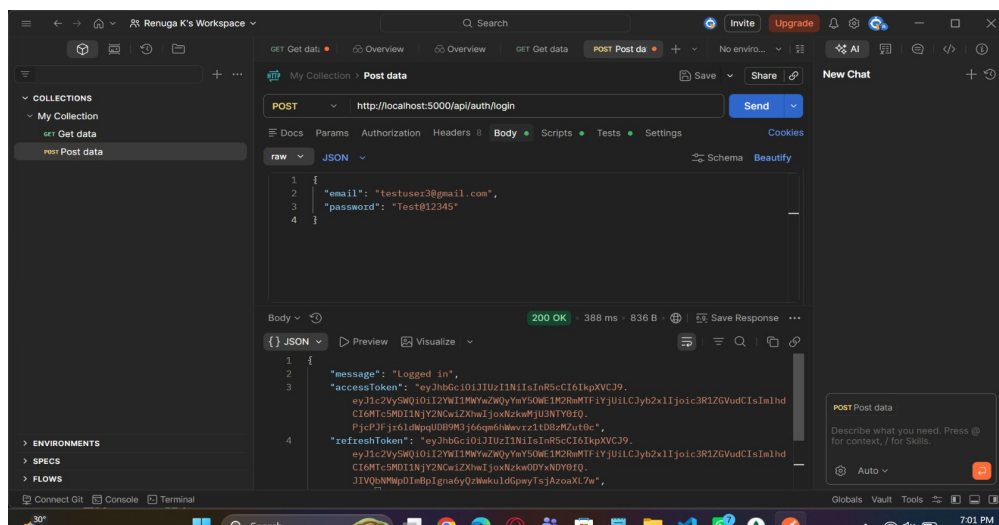
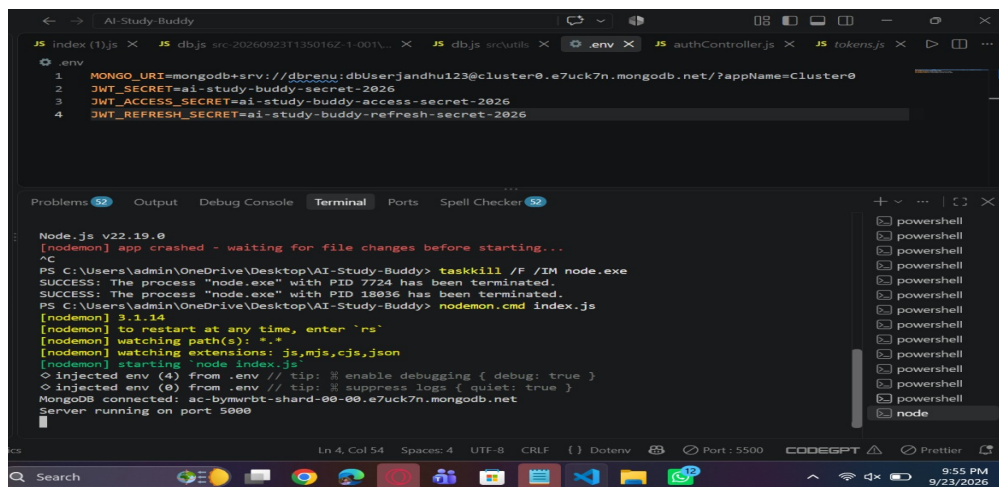
2. Anthropic Claude API Key

- Go to the Anthropic Console.
- Create a free account or sign in.
- Navigate to **API Keys** in the left sidebar and click **Create Key**.
- Copy the key immediately and save it, as it will not be shown again

3. OpenAI API Key

- Log in to your [OpenAI account](#).
- Go to your API keys section and click **Create new secret key**.
- Save the key securely and assign it to the OPENAI_API_KEY environment variable

StudyBuddy



Study Buddy (One-on-One)

A study buddy is best for deep focus, personalized pacing, and consistent, quiet accountability.

➤ Pros:

- **Fewer distractions:** Easier to stay on task with only two people than a larger crowd.
- **Deep focus:** Better for tackling complex or brand-new material where you need quiet time to read and think.
- **Flexible scheduling:** Much easier to coordinate schedules with just one other person.
- **Stronger emotional support:** Easier to build rapport, share stress, and match each other's pace.

➤ Cons:

- **Limited perspectives:** You only bounce ideas off one other brain.
- **Risk of stalling:** If both of you get stuck on a difficult concept, you might not find the answer easily.

Study Group (Three or More People)

A study group is best for reviewing material, dividing large workloads, and hearing different viewpoints.

➤ Pros:

- **Diverse strengths:** Different people understand different parts of the syllabus and can teach what they know.
- **Divide and conquer:** Great for splitting up large sets of study notes or research topics.
- **Great for auditory learners:** Excellent for debating concepts, quizzing each other out loud, and talking through problems.

➤ Cons:

- **High distraction risk:** Groups easily slip into chatting and socializing if not strictly disciplined.
- **Scheduling conflicts:** Harder to find a time and place that works for everyone.

You can find a study partner online through dedicated matching platforms like StudyBuddy or global focus communities like Study Together.

➤ **Where to Look**

- **In class:** Talk to the person sitting next to you and ask for their contact info to share notes or quiz each other.
- **School portals:** Use platforms like Blackboard or campus Facebook groups to message classmates and form a group.
- **Online matching platforms:** Use dedicated matching sites like MoocLab to find peers based on your specific subjects and goals.

➤ **Tips for Success**

- **Set clear rules:** Agree on a schedule, limit phone use, and use focus tools like pomodoro timers
- **Choose the right setup:** Decide if you want an active quiz partner or a silent "body-doubling" partner for quiet accountability.